

# Electronic Devices

## Final Term Lecture - 02

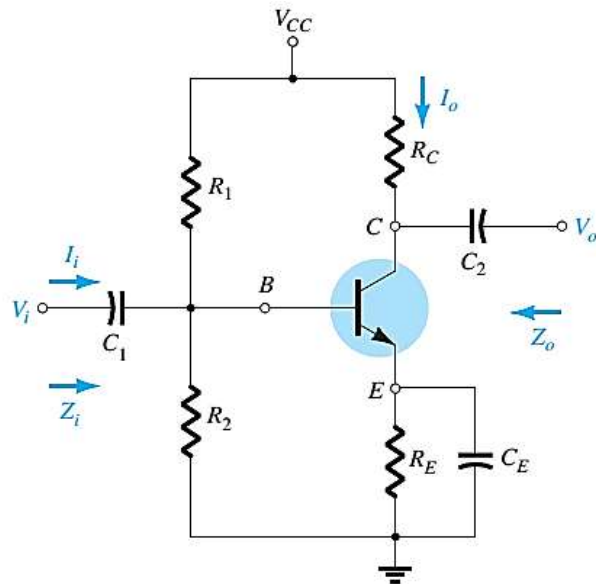
Reference book:

**Electronic Devices and Circuit Theory  
(Chapter-5)**

Robert L. Boylestad and L. Nashelsky , (11<sup>th</sup> Edition)

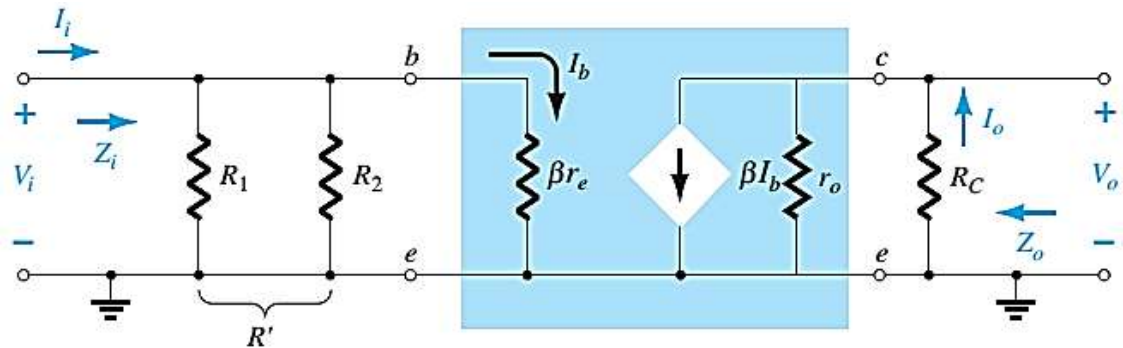


# COMMON-EMITTER VOLTAGE-DIVIDER BIAS



**FIG. 5.26**

Voltage-divider bias configuration.



**FIG. 5.27**

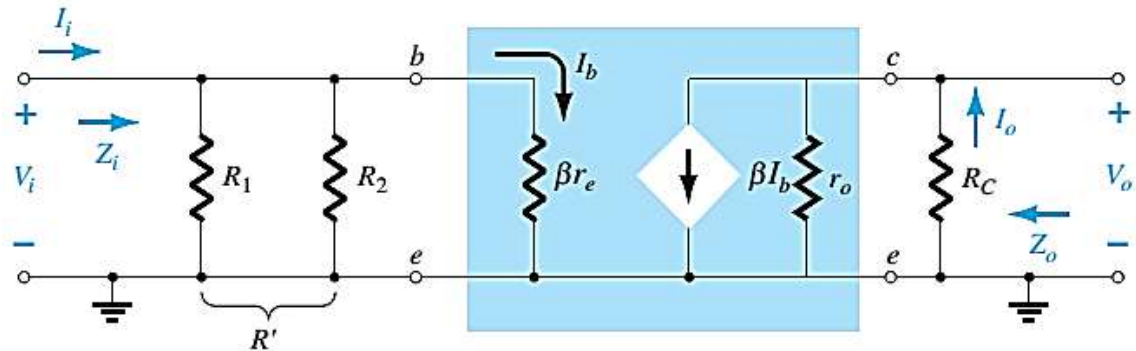
Substituting the  $r_e$  equivalent circuit into the ac equivalent network of Fig. 5.26.

# COMMON-EMITTER VOLTAGE-DIVIDER BIAS

## INPUT IMPEDANCE, $Z_i$

$$R' = R_1 \parallel R_2$$

$$Z_i = R' \parallel \beta r_e$$



## OUTPUT IMPEDANCE, $Z_o$

$$Z_o = R_C \parallel r_o$$

$$Z_o \cong R_C \mid_{r_o \geq 10R_C}$$

## VOLTAGE GAIN, $A_v$

$$V_o = -\beta I_b (R_C \parallel r_o) = -\beta \left( \frac{V_i}{\beta r_e} \right) (R_C \parallel r_o); I_b = \frac{V_i}{\beta r_e}$$

$$A_v = \frac{V_o}{V_i} = -\frac{(R_C \parallel r_o)}{r_e}, A_v = -\frac{R_C}{r_e} \mid_{r_o \geq 10R_C}$$

# EXAMPLE

- **EXAMPLE 5.2:** For the network of Fig. 5.28 :
- Determine  $r_e$ ,  $Z_i$ ,  $Z_o$  (with  $r_o = \infty$ ),  $A_v$  (with  $r_o = \infty$ ) and Repeat with  $r_o = 50 \text{ k}\Omega$ .

a. DC: Testing  $\beta R_E > 10R_2$ ,

$$(90)(1.5 \text{ k}\Omega) > 10(8.2 \text{ k}\Omega)$$

$$135 \text{ k}\Omega > 82 \text{ k}\Omega \text{ (satisfied)}$$

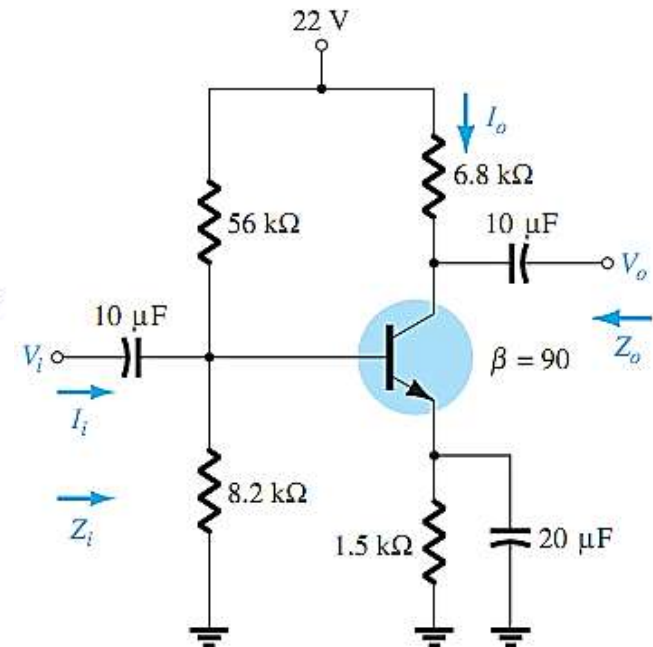
Using the approximate approach, we obtain

$$V_B = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{(8.2 \text{ k}\Omega)(22 \text{ V})}{56 \text{ k}\Omega + 8.2 \text{ k}\Omega} = 2.81 \text{ V}$$

$$V_E = V_B - V_{BE} = 2.81 \text{ V} - 0.7 \text{ V} = 2.11 \text{ V}$$

$$I_E = \frac{V_E}{R_E} = \frac{2.11 \text{ V}}{1.5 \text{ k}\Omega} = 1.41 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.41 \text{ mA}} = 18.44 \text{ }\Omega$$



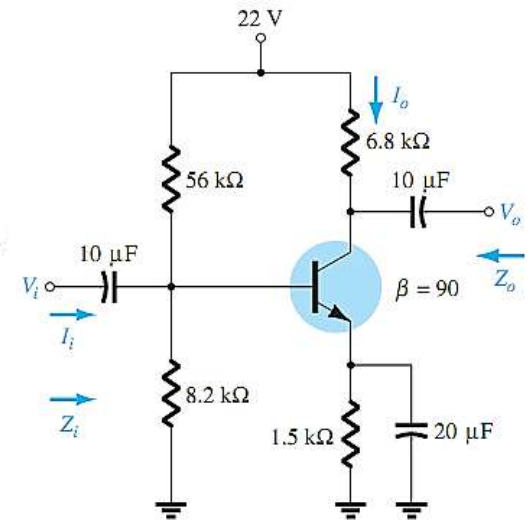
**FIG. 5.28**

Example 5.2.

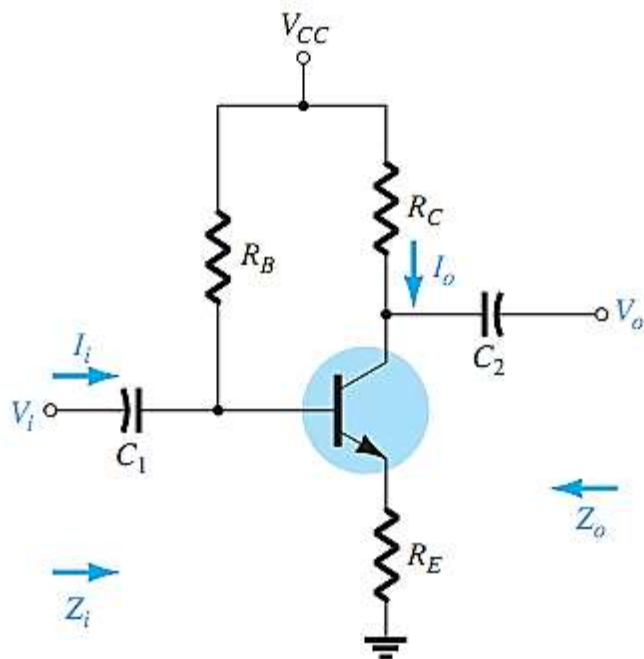
## EXAMPLE Contd.

- b.  $R' = R_1 \parallel R_2 = (56 \text{ k}\Omega) \parallel (8.2 \text{ k}\Omega) = 7.15 \text{ k}\Omega$   
 $Z_i = R' \parallel \beta r_e = 7.15 \text{ k}\Omega \parallel (90)(18.44 \text{ }\Omega) = 7.15 \text{ k}\Omega \parallel 1.66 \text{ k}\Omega$   
 $= 1.35 \text{ k}\Omega$
- c.  $Z_o = R_C = 6.8 \text{ k}\Omega$
- d.  $A_v = -\frac{R_C}{r_e} = -\frac{6.8 \text{ k}\Omega}{18.44 \text{ }\Omega} = -368.76$
- e.  $Z_i = 1.35 \text{ k}\Omega$   
 $Z_o = R_C \parallel r_o = 6.8 \text{ k}\Omega \parallel 50 \text{ k}\Omega = 5.98 \text{ k}\Omega \text{ vs. } 6.8 \text{ k}\Omega$   
 $A_v = -\frac{R_C \parallel r_o}{r_e} = -\frac{5.98 \text{ k}\Omega}{18.44 \text{ }\Omega} = -324.3 \text{ vs. } -368.76$

There was a measurable difference in the results for  $Z_o$  and  $A_v$ , because the condition  $r_o \geq 10R_C$  was *not* satisfied.

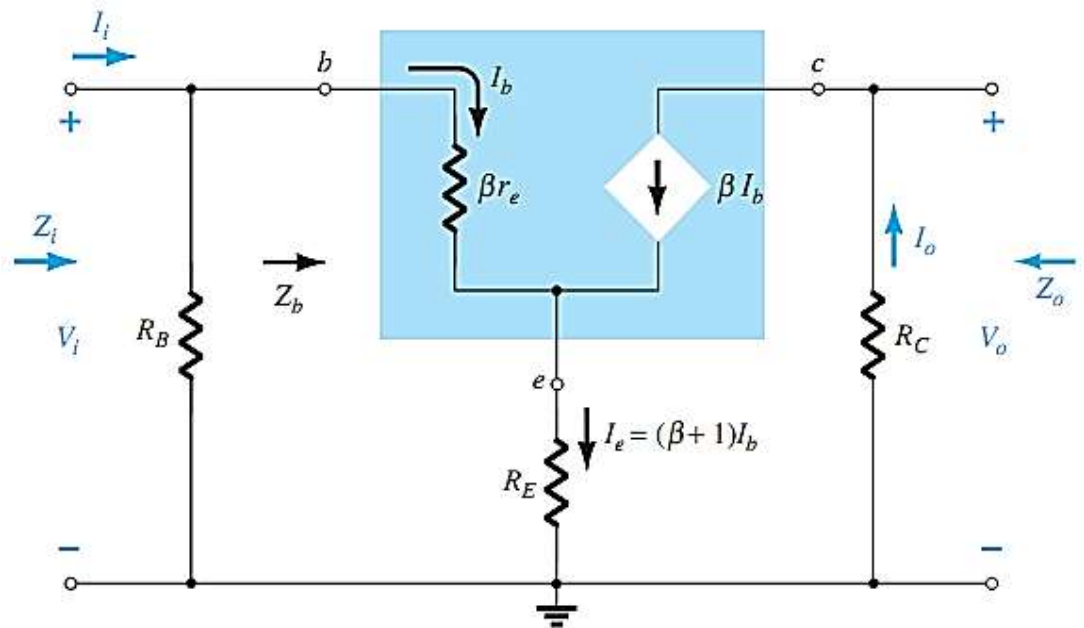


## COMMON-EMITTER EMITTER-BIAS CONFIGURATION: UNBYPASSED $R_E$



**FIG. 5.29**

*CE emitter-bias configuration.*



**FIG. 5.30**

*Substituting the  $r_e$  equivalent circuit into the ac equivalent network of Fig. 5.29.*

# IMPEDANCE CALCULATION

## INPUT IMPEDANCE, $Z_i$

$$V_i = I_b \beta r_e + I_e R_E$$

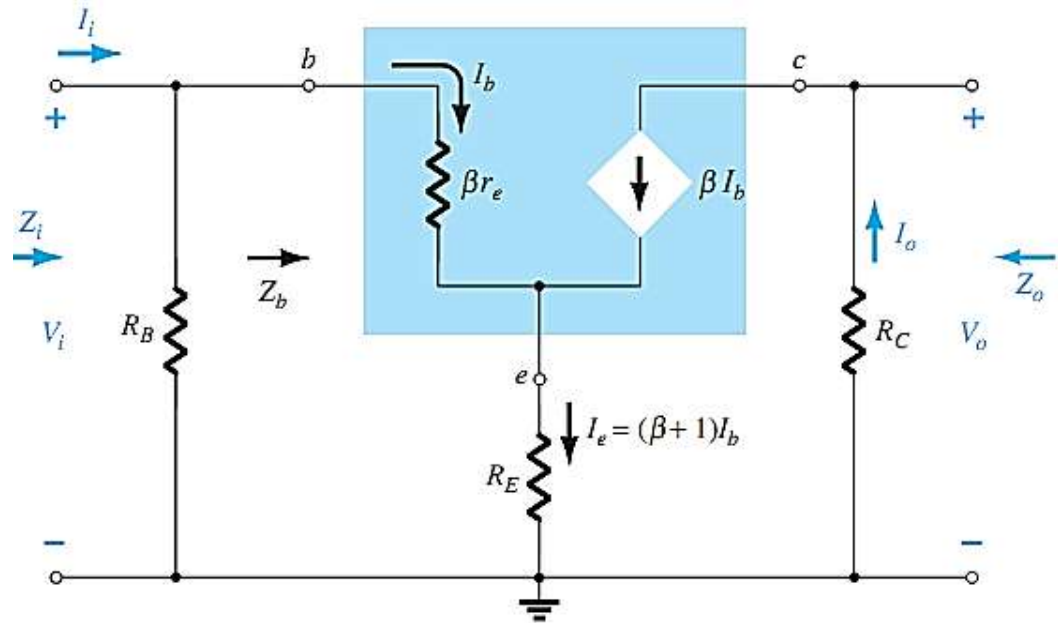
$$= I_b \beta r_e + (\beta + 1) I_b R_E$$

$$Z_b = \frac{V_i}{I_b} = \beta r_e + (\beta + 1) R_E$$

$$Z_b \cong \beta r_e + \beta R_E = \beta(r_e + R_E)$$

$$Z_b \cong \beta R_E \quad \text{for } R_E \gg r_e$$

$$Z_i = R_B || Z_b$$



## OUTPUT IMPEDANCE, $Z_o$

$$Z_o = R_C$$

# GAIN CALCULATIONS

## VOLTAGE GAIN, $A_v$

$$V_o = -I_o R_C = -\beta I_b R_C = -\beta \left( \frac{V_i}{Z_b} \right) R_C$$

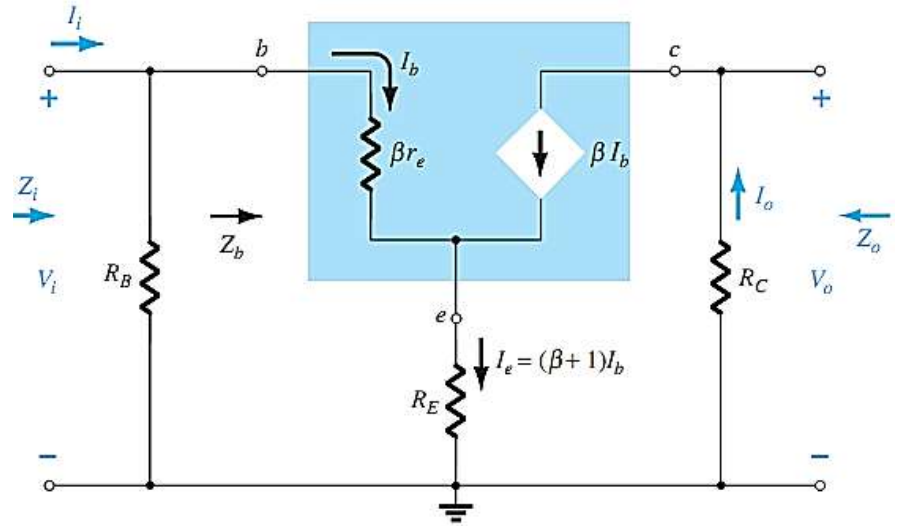
$$A_v = \frac{V_o}{V_i} = -\frac{\beta R_C}{Z_b}$$

*Substituting*  $Z_b \cong \beta(r_e + R_E)$

$$A_v = \frac{V_o}{V_i} = -\frac{R_C}{r_e + R_E}$$

*For the approximation*  $Z_b \cong \beta R_E$

$$A_v = \frac{V_o}{V_i} = -\frac{R_C}{R_E}$$



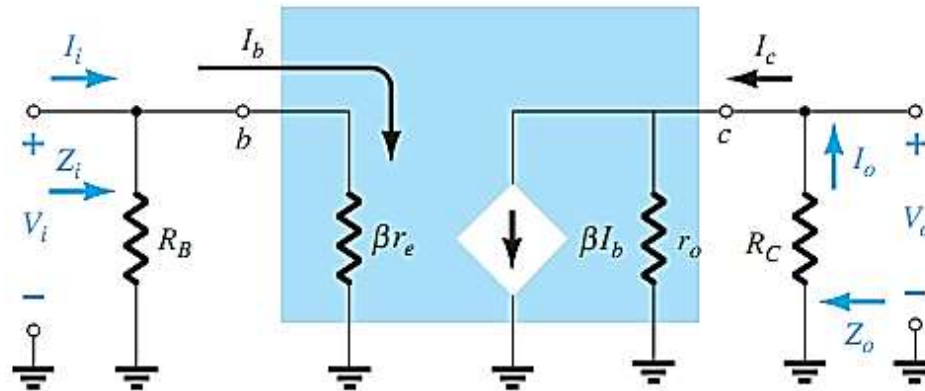
The **negative sign** in gain equations reveals **180° phase shift** between input and output waveforms.



## COMMON-EMITTER EMITTER-BIAS CONFIGURATION: BYPASSED $R_E$

### Bypassed

If  $R_E$  is bypassed by an emitter capacitor  $C_E$ , the complete  $r_e$  equivalent model can be substituted, resulting in the same equivalent network as Fig. 5.22. Equations of slide no. 13 are therefore applicable.



**FIG. 5.22**

*Substituting the  $r_e$  model into the network of Fig. 5.21.*

# EXAMPLE

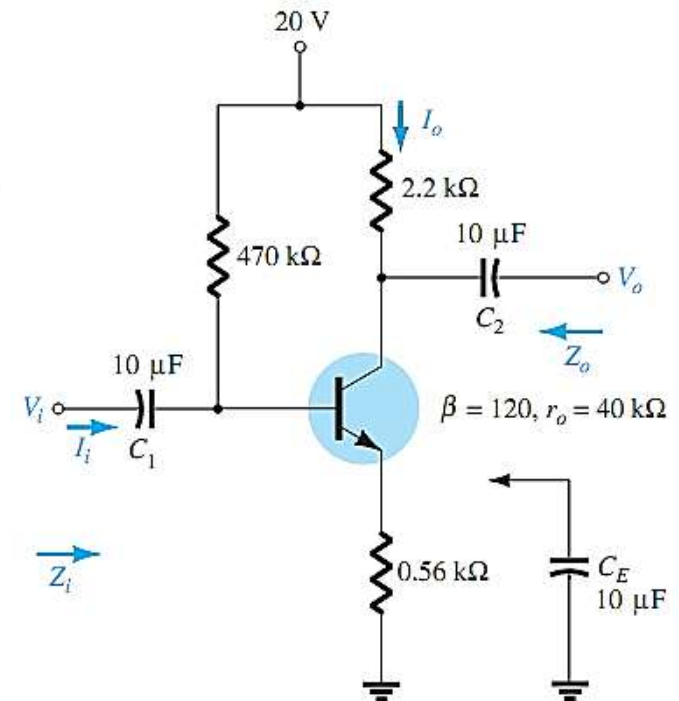
- EXAMPLE 5.3:** For the network of following Fig, without  $C_E$  (unbypassed), determine:  $r_e$ ,  $Z_i$ ,  $Z_o$  &  $A_v$ .

a. DC:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + (121)0.56 \text{ k}\Omega} = 35.89 \mu\text{A}$$

$$I_E = (\beta + 1)I_B = (121)(35.89 \mu\text{A}) = 4.34 \text{ mA}$$

and  $r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{4.34 \text{ mA}} = \mathbf{5.99 \Omega}$



**FIG. 5.32**  
Example 5.3.

## EXAMPLE Contd.

b. Testing the condition  $r_o \geq 10(R_C + R_E)$ , we obtain

$$40 \text{ k}\Omega \geq 10(2.2 \text{ k}\Omega + 0.56 \text{ k}\Omega)$$

$$40 \text{ k}\Omega \geq 10(2.76 \text{ k}\Omega) = 27.6 \text{ k}\Omega \text{ (satisfied)}$$

Therefore,

$$\begin{aligned} Z_b &\cong \beta(r_e + R_E) = 120(5.99 \text{ }\Omega + 560 \text{ }\Omega) \\ &= 67.92 \text{ k}\Omega \end{aligned}$$

and

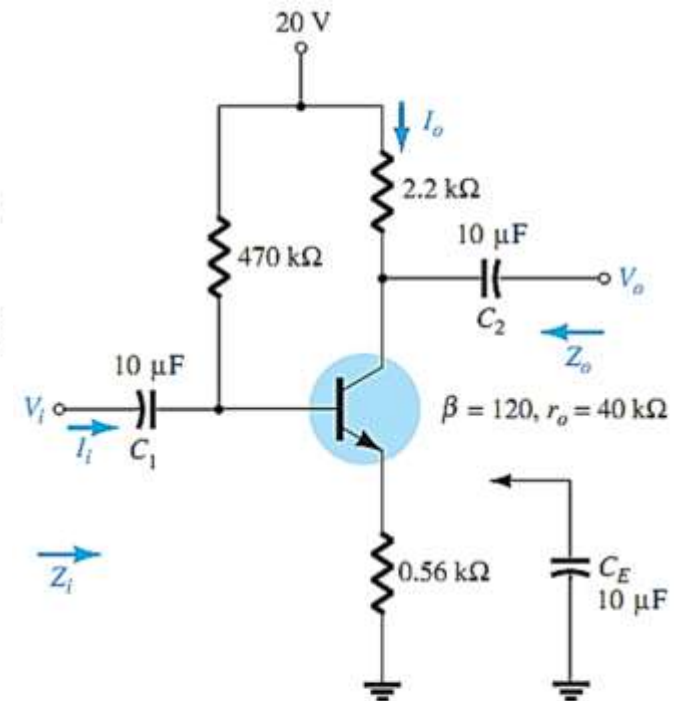
$$\begin{aligned} Z_i &= R_B \parallel Z_b = 470 \text{ k}\Omega \parallel 67.92 \text{ k}\Omega \\ &= \mathbf{59.34 \text{ k}\Omega} \end{aligned}$$

c.  $Z_o = R_C = \mathbf{2.2 \text{ k}\Omega}$

d.  $r_o \geq 10R_C$  is satisfied. Therefore,

$$\begin{aligned} A_v &= \frac{V_o}{V_i} \cong -\frac{\beta R_C}{Z_b} = -\frac{(120)(2.2 \text{ k}\Omega)}{67.92 \text{ k}\Omega} \\ &= \mathbf{-3.89} \end{aligned}$$

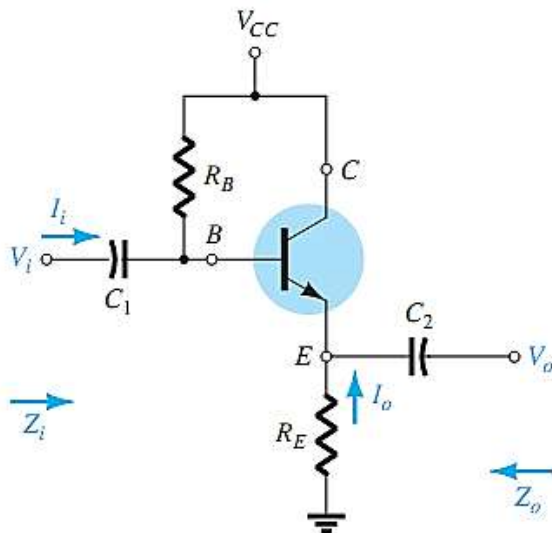
compared to  $-3.93$  using Eq. (5.20):  $A_v \cong -R_C/R_E$ .



**FIG. 5.32**

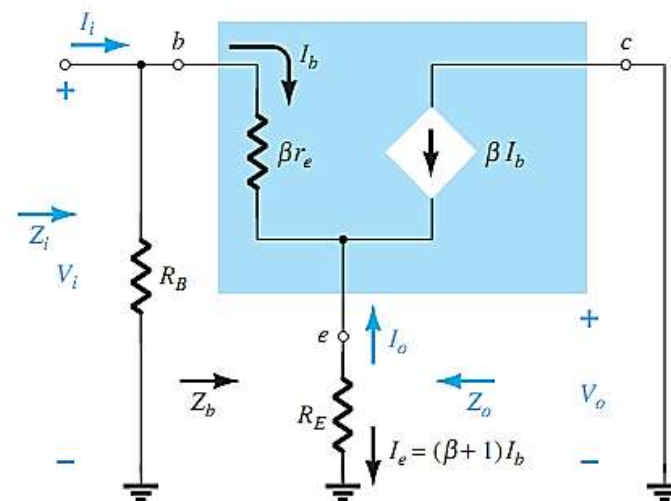
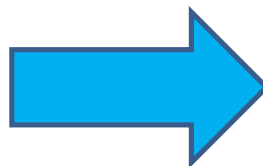
Example 5.3.

# EMITTER-FOLLOWER CONFIGURATION



**FIG. 5.36**

*Emitter-follower configuration.*



**FIG. 5.37**

*Substituting the  $r_e$  equivalent circuit into the ac equivalent network of Fig. 5.36.*

- This is also known as the common-collector configuration.
- The input is applied to the base and the output is taken from the emitter.
- There is no phase shift between input and output.

# IMPEDANCE CALCULATIONS

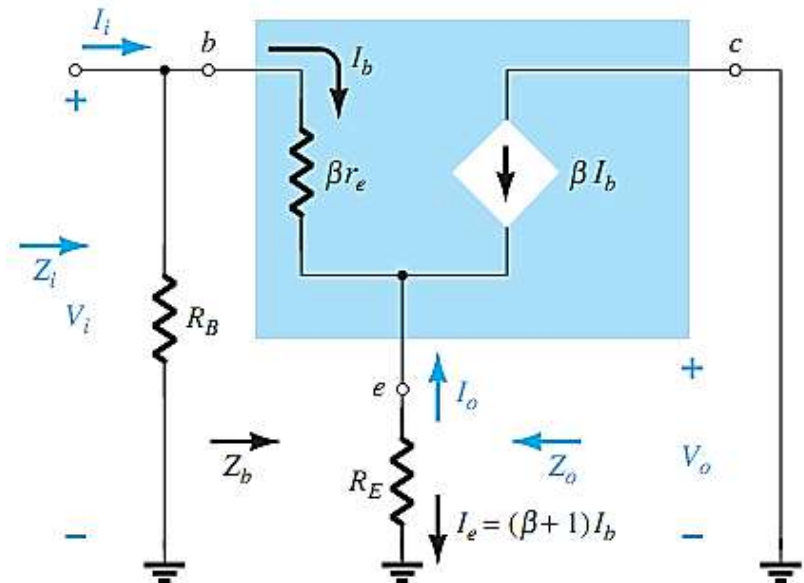
## INPUT IMPEDANCE, $Z_i$

$$Z_i = R_B || Z_b$$

$$Z_b = \beta r_e + (\beta + 1)R_E$$

$$Z_b \cong \beta(r_e + R_E)$$

$$Z_b \cong \beta R_E \quad \text{for } R_E \gg r_e$$



# IMPEDANCE CALCULATIONS

## OUTPUT IMPEDANCE, $Z_o$

$$I_b = \frac{V_i}{Z_b},$$

$$I_E = (\beta + 1)I_b = (\beta + 1) \frac{V_i}{Z_b}$$

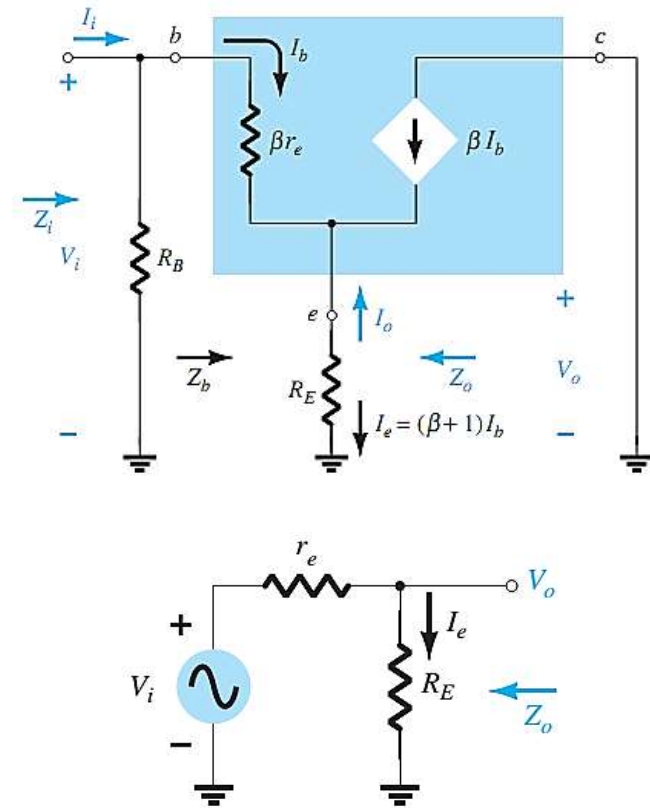
$$I_E = \frac{(\beta + 1)V_i}{\beta r_e + (\beta + 1)R_E}$$

Since  $(\beta + 1) \cong \beta$

$$I_E = \frac{V_i}{r_e + R_E}$$

To determine  $Z_o$ ,  $V_i$  is set to zero

$$Z_o = R_E || r_e \quad Z_o = r_e \mid_{R_E \gg r_e}$$



**FIG. 5.38**

Defining the output impedance for the emitter-follower configuration.

# GAIN CALCULATIONS

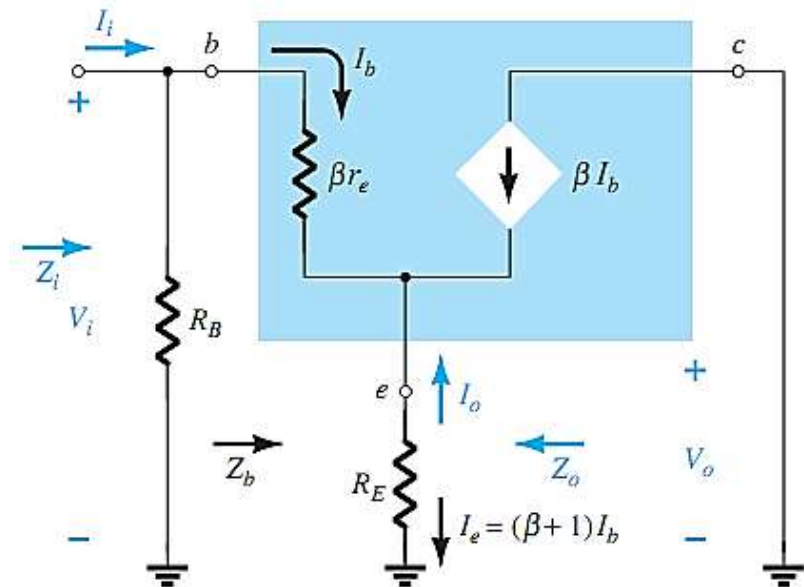
## VOLTAGE GAIN, $A_v$

$$V_o = I_e R_E = (\beta + 1) I_b R_E = \frac{V_i (\beta + 1) R_E}{Z_b}$$

when  $r_o \geq 10 R_E$  and  $\beta + 1 \cong \beta$

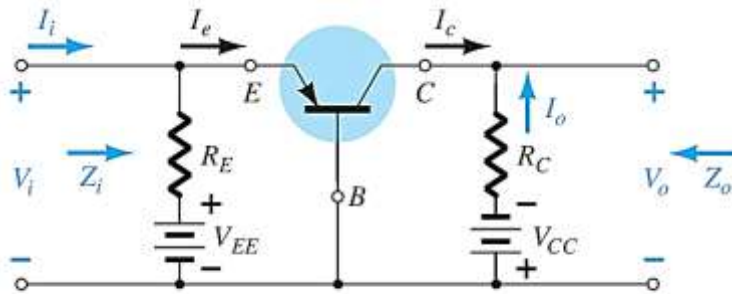
But  $Z_b \cong \beta(r_e + R_E)$

$$A_v = \frac{V_o}{V_i} \cong \frac{\beta R_E}{\beta(r_e + R_E)} \cong \frac{R_E}{(r_e + R_E)}$$



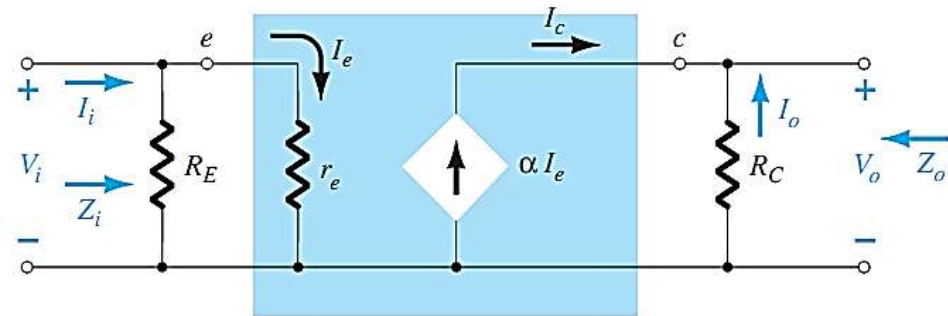
□ See Example 5.7

# COMMON-BASE CONFIGURATION



**FIG. 5.42**

*Common-base configuration.*



**FIG. 5.43**

*Substituting the  $r_e$  equivalent circuit into the ac equivalent network of Fig. 5.44.*

- The input is applied to the emitter.
- The output is taken from the collector.
- Low input impedance.
- High output impedance.
- Very high voltage gain.
- No phase shift between input and output.



# CALCULATIONS

## INPUT IMPEDANCE, $Z_i$

$$Z_i = R_E \parallel r_e$$

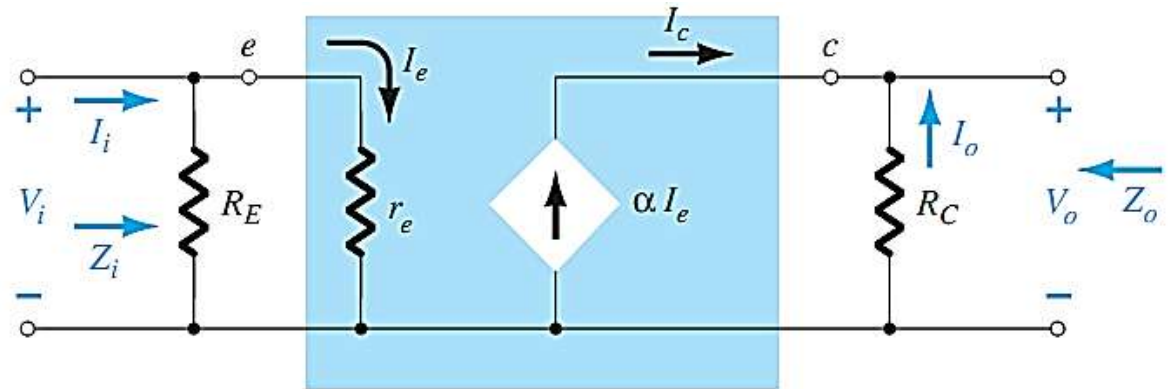
## OUTPUT IMPEDANCE, $Z_o$

$$Z_o = R_C$$

## VOLTAGE GAIN, $A_v$

$$V_o = -I_o R_C = -(I_C) R_C = \alpha I_e R_C; \quad I_e = \frac{V_i}{r_e}$$

$$V_o = \alpha \left( \frac{V_i}{r_e} \right) R_C; \quad A_v = \frac{V_o}{V_i} = \frac{\alpha R_C}{r_e} \cong \frac{R_C}{r_e}$$



## CURRENT GAIN, $A_i$

Assuming  $R_E \gg r_e$

$$I_e = I_i$$

$$I_o = -\alpha I_e = -\alpha I_i$$

$$A_i = \frac{I_o}{I_i} = -\alpha \cong -1$$

# COMMON-BASE CONFIGURATION

- **Phase Relationship:**

The fact that  $A_v$  is a positive number shows that  $V_o$  and  $V_i$  are in phase for the common-base configuration.

- **Effect of  $r_o$ :**

For the common-base configuration,  $r_o = 1/h_{ob}$  is typically in the megohm range and sufficiently larger than the parallel resistance  $R_C$  to permit the approximation  $r_o \parallel R_C \cong R_C$ .

## EXAMPLE

- EXAMPLE 5.8:** For the network of following figure, determine:  $r_e$ ,  $Z_i$ ,  $Z_o$ ,  $A_v$ ,  $A_i$ .

$$\text{a. } I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{2 \text{ V} - 0.7 \text{ V}}{1 \text{ k}\Omega} = \frac{1.3 \text{ V}}{1 \text{ k}\Omega} = 1.3 \text{ mA}$$

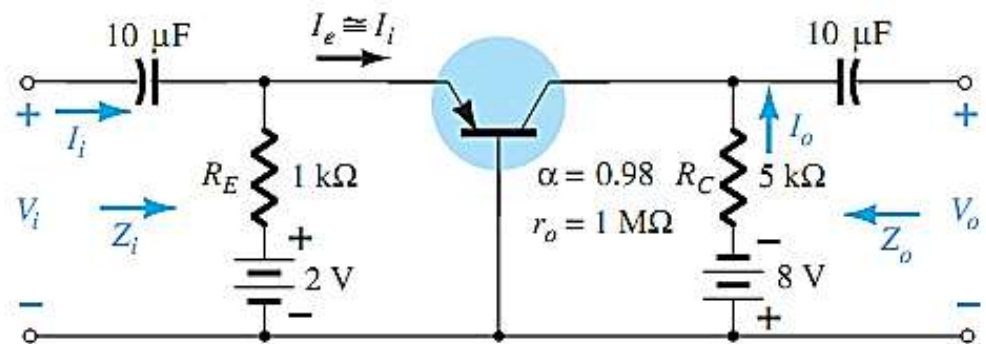
$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.3 \text{ mA}} = \mathbf{20 \text{ }\Omega}$$

$$\text{b. } Z_i = R_E \parallel r_e = 1 \text{ k}\Omega \parallel 20 \text{ }\Omega \\ = \mathbf{19.61 \text{ }\Omega} \cong r_e$$

$$\text{c. } Z_o = R_C = \mathbf{5 \text{ k}\Omega}$$

$$\text{d. } A_v \cong \frac{R_C}{r_e} = \frac{5 \text{ k}\Omega}{20 \text{ }\Omega} = \mathbf{250}$$

$$\text{e. } A_i = \mathbf{-0.98} \cong -1$$



**FIG. 5.44**

Example 5.8.

*End of Lecture-2*